ShopForMe

A succinct description of your project :

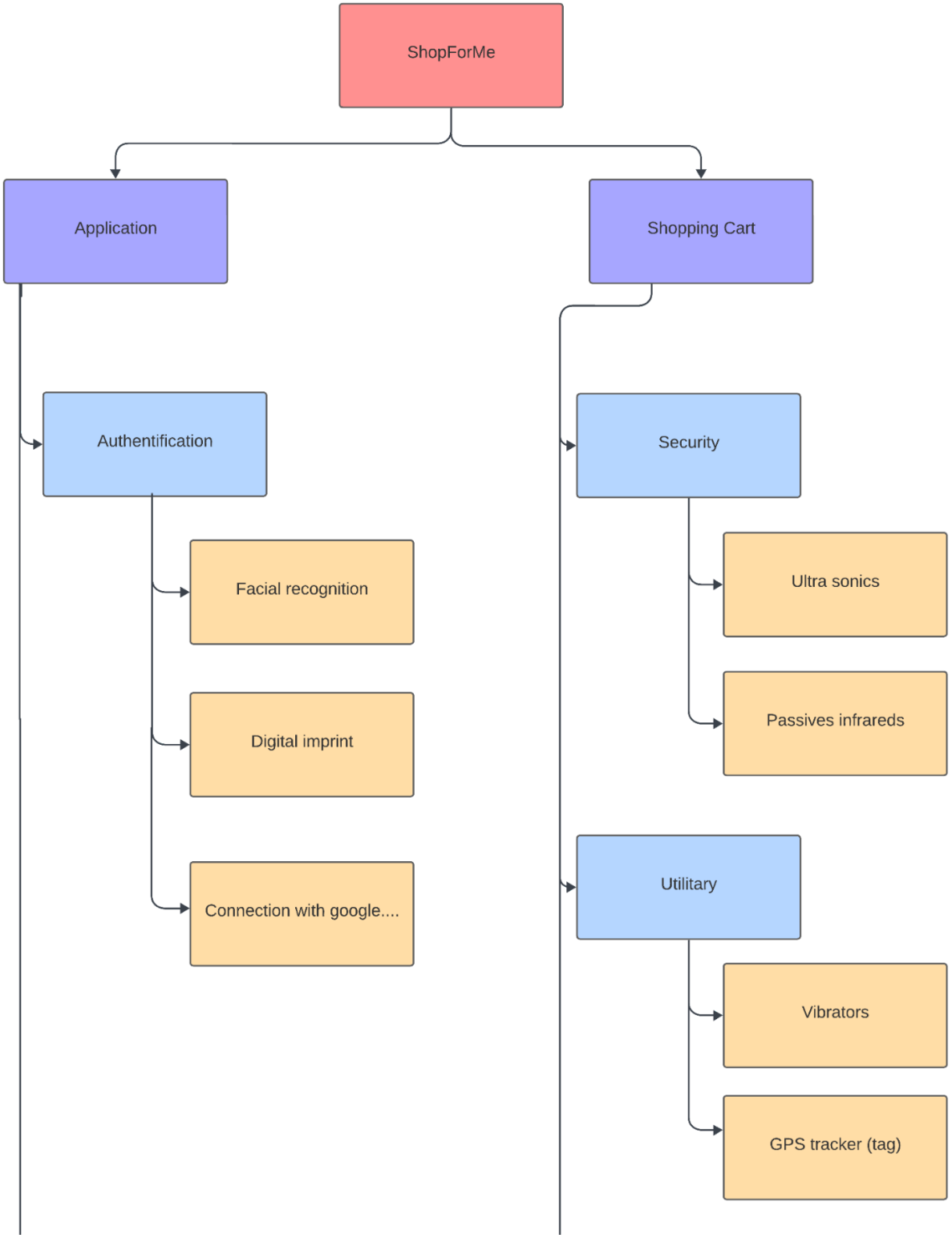
Français :

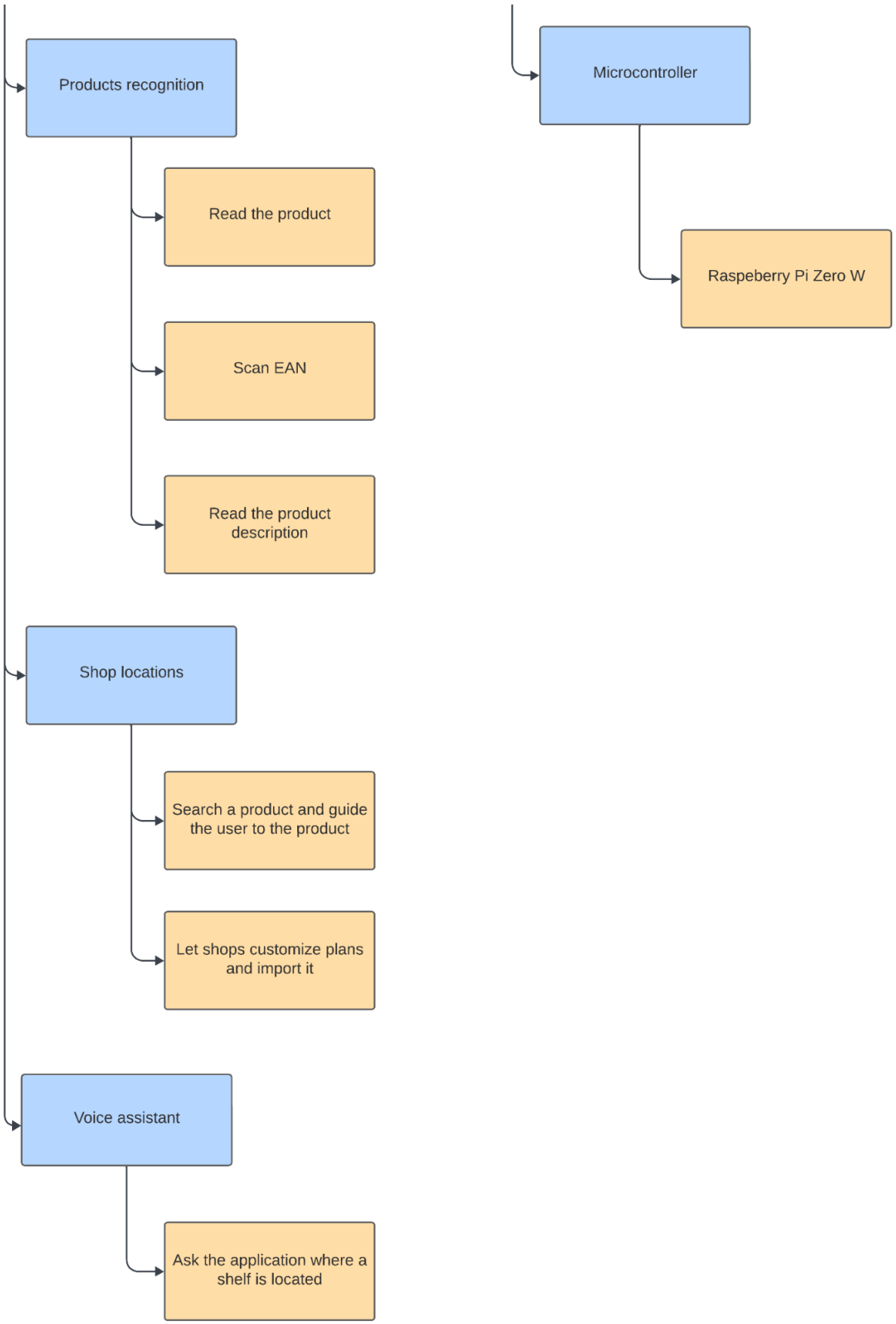
La mise en place d'une application permettant d'aider les malvoyants à faire leurs courses. L'application pourra guider l'utilisateur grâce à la caméra et la reconnaissance de l'environnement autour de lui grâce à l'intelligence artificielle.
Elle devra fonctionner de paire avec les objets connectés du chariot qui se charge de la sécurité du malvoyant et des personnes environnantes.

English :

The implementation of an application to assist visually impaired individuals with their shopping. The application will be able to guide the user through the camera and environmental recognition around them using artificial intelligence. It will need to work in conjunction with the connected objects on the cart, which ensures the safety of the visually impaired person and those in the surrounding area.

Project Breakdown Structure (PBS):




Products recognition
Read the product
Scan EAN
Read the product description
Shop locations
Search a product and guide the user to the product
Let shops customize plans and import it
Voice assistant
Ask the application where a shelf is located
Microcontroller
Raspeberry Pi Zero W

For the security of the person using the cart, we add passive infrared and ultrasonics sensors. They both have one different objective, for infrared sensors it is to identify if someone is close to the cart and for ultrasonics ones, it is to identify if the blind user of the cart is close to a shelf.

Working with sensors, the cart is composed of two vibrators which prevents obstacles by notifying the user. For example : obstacles are on the left, and the left vibrator vibrates.

ultraSonic

The GPS tracker is used to guide the user in the shop using plans referenced in the application. The tracker is included in the cart.

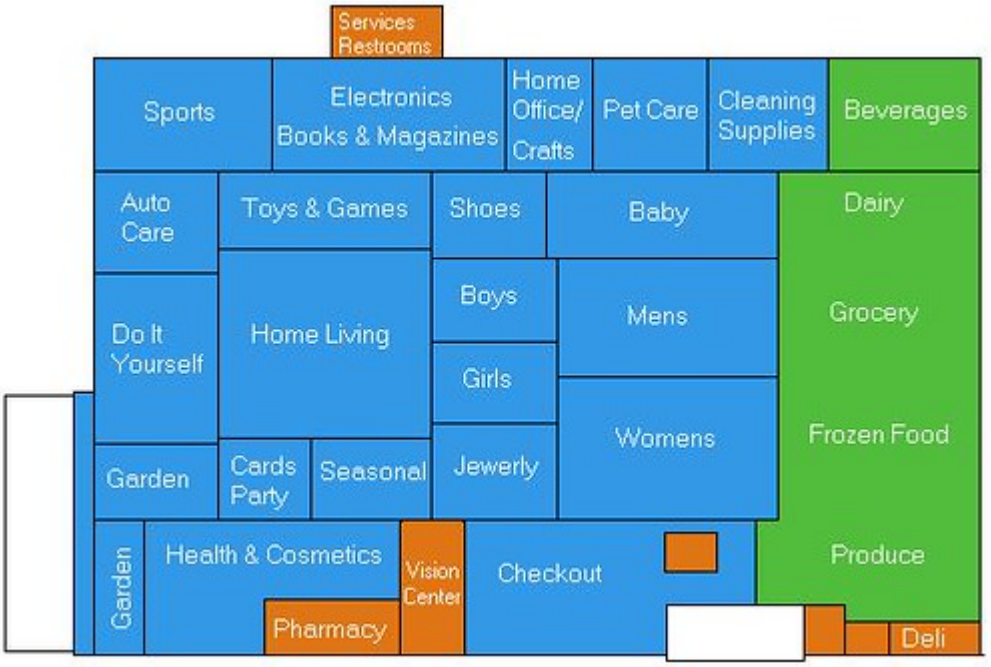
For the shopping cart we can maybe use the raspberry for additional features.

To connect to the application a user can authenticate with a facial recognition, digital imprint or we can imagine a connection using google or just by using his application account.

To help blind people, in the application, it is possible to use the camera as a friend. Cameras are the eyes of the friend and the microphone of the mobile is his mouth. The application will tell information about the product using this process.

If some more details about the product are needed, users can scan the EAN (European Article Numbering), and some more details can be given to them such as the composition, nutrients etc.

To help in depth blind people, it has already been mentioned that they have to be guided to their product. In order to do this, grocery shopping teams have to give a plan of their shop.



A voice assistant is of course accessible all over the application.

Powerpoint presentation :

https://docs.google.com/presentation/d/1GQ-TzM-H8auxtV48QGTGt2P7-miN2VkisHw5iMFVSmE/edit?usp=sharing

ShopForMe

FRANQUES Matthieu
HADDOUCH Elhoucine
LE DOUARIN Frédéric

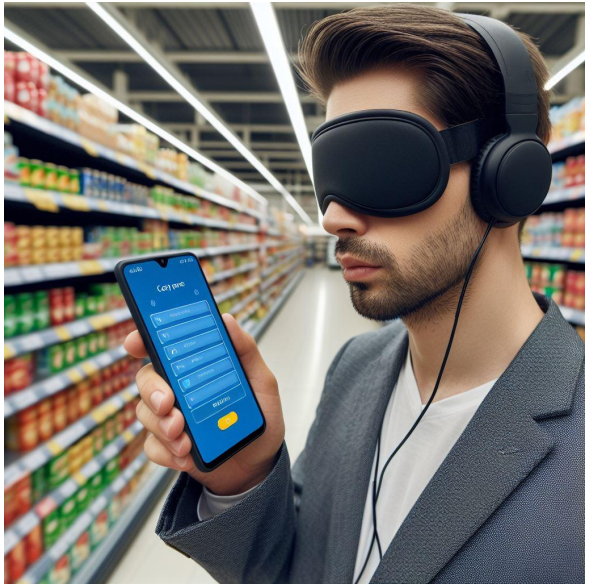
Facts

- Global blindness set to triple by 2050
- 115 million blind people by 2050
- 550 million people with sight problems by 2050

What is ShopForMe ?

Description of ShopForMe: application

- Allow blind people to do their grocery shopping
- Guide users using their phone's camera and voice assistance to tell them what they are looking for
- Recognizing the environment around with artificial intelligence and identify products



Description of ShopForMe: shopping cart

- Guide and notify the user of possible collisions
- Possibility to plug your mobile to the cart for additional features
- Differents sizes to fit in the size of the shop

Typicals examples of the shopping cart

- Detects shop shelves and humans

- You can find out where the items you're looking for are

A danger for the blind person and others ?

- Slight vibrations on the handlebars to let you know if a human or a shelf is in the way

- A shelf on the right? A vibration on the right

  A human opposite? A vibration across the entire handlebar

Arrived at the right shelf, then what ?

- Blind people can look up and down with their phones, and get an idea of the product thanks to the camera

- AI will make it possible to identify the product and the blind person will be able to hear its description

Thanks for listening!